

LESSON 2-2 • HOMEWORK PACKET • SHEET 1 OF 2

Building a Histogram

Maryland Standard 6.SP.B.4 • Learning target – I can group data into equal intervals and build a histogram from a frequency table.

REMEMBER

- A **histogram** groups data into **equal intervals**, like 0-9, 10-19, 20-29.
- The **frequency** is how many values fall in an interval. It sets the bar height.
- Intervals do not overlap, and the bars **touch**.

WORKED EXAMPLE

Minutes of practice: 4, 15, 22, 9, 18, 27, 13, 16, 21, 11.
Make a frequency table.

Step 1 0-9: 4, 9, so the frequency is 2.

Step 2 10-19: 15, 18, 13, 16, 11, so it is 5.

Step 3 20-29: 22, 27, 21, so it is 3.

Answer 2, 5, 3

PRACTICE

show your work • label units

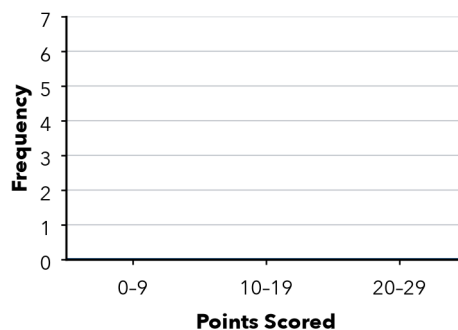
1. **Points Scored.** A player scored these points in 12 games: 8, 14, 21, 5, 17, 12, 25, 19, 11, 23, 16, 9. Complete the frequency table.

Points	Tally	Frequency
0-9		
10-19		
20-29		

2. Draw the histogram for Problem 1. Bars should touch.

a. Which interval has the tallest bar? _____

b. Games with 10 or more points: _____



3. **Picnic Ages.** 34, 22, 41, 38, 27, 45, 31, 36, 29, 48 years. Find each frequency.

a. 20-29: _____

b. 30-39: _____

c. 40-49: _____

4. **Missing Bar.** A histogram shows the long-jump distances of 24 students. Three bars have heights 5, 9, and 3. There is one more bar. How tall is it?

Answer _____ students

5. **Find the Error.** Marcus used the intervals 0-9, 10-14, and 15-29. Explain his mistake and fix it.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Ask family members how many minutes they read today. Group the answers in intervals of 10.

LESSON 2-2 • HOMEWORK PACKET • SHEET 2 OF 2**Describing Histogram Shape**

Maryland Standard 6.SP.B.4 • Learning target – I can describe the shape of a histogram using its peak, spread, and balance.

REMEMBER

- **Peak:** the tallest bar. **Symmetric:** sides match.
- **Skewed right:** the tail stretches right. **Skewed left:** the tail stretches left.
- A histogram shows counts, not exact values.

WORKED EXAMPLE

Bars of height 2, 7, 4, 2, 1 over 0-9, 10-19, 20-29, 30-39, 40-49. Shape?

Step 1 The peak is 10-19, near the left.

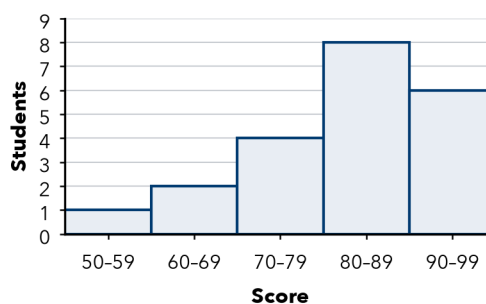
Step 2 The bars shrink to the right: a tail on the right.

Answer Skewed right

PRACTICE

show your work • label units

- 1. Test Scores.** The histogram shows the scores on a class test.



- a. Number of students: _____
- b. Peak interval: _____
- c. Shape: _____
- d. Scores of 80 or more: _____

- 2.** Bar heights are listed from left to right. Name each shape.

- a. 1, 4, 8, 4, 1 _____ b. 9, 6, 3, 2, 1 _____ c. 1, 2, 3, 6, 10 _____
- d. 3, 5, 9, 5, 3 _____ e. 12, 8, 5, 3, 2 _____ f. 2, 3, 5, 8, 12 _____

- 3. Race Times.** Bars of 3, 8, 11, 6, 2 show 30 runners' times: 20-24, 25-29, 30-34, 35-39, 40-44 min.

- a. How many runners finished in less than 30 minutes? _____
- b. How many finished in 35 minutes or more? _____

- 4. Stretch.** Write 5 bar heights, from left to right, for a histogram of **20** values that is **skewed left**.

Bar heights _____

- 5. Explain.** Ramon says the Test Scores histogram shows a score of 85. Is he right? Explain.

TEST PRACTICE

choose the best answer

- 6.** Bar heights are listed from left to right. Which histogram is skewed right?

- A** 2, 5, 9, 5, 2 **B** 10, 7, 4, 2, 1 **C** 1, 2, 4, 7, 10 **D** 5, 5, 5, 5, 5

- 7.** Bars of height 4, 7, and 5 sit over 0-9, 10-19, and 20-29. How many values are 10 or more?

- A** 7 **B** 11 **C** 12 **D** 16

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____